8. Conclusion

In this study, we introduce graphic-form games and preference graphic-form games as intuitive reformulations of cooperative games. These reformulations can effectively evaluate and pinpoint cooperative incompatibility during the learning process of Zero-Shot Coordination (ZSC) algorithms, thereby further addressing the issue of cooperative incompatibility in zero-shot human-AI coordination. Additionally, we propose the COLE framework, designed to iteratively approximate the best response preferred by teammates within the latest population. Theoretically, we provide proof that COLE framework converges towards the locally optimal strategy preferred by the rest of the population. If the in-degree centrality is selected as the preference centrality function, the convergence rate would achieve Q-sublinear status.

We also implemented an online pipeline for the Overcooked Human-AI experiment, which allows for easy modifications of questionnaires, model weights, and other elements. To the best of our knowledge, it is the first comprehensive human-AI experimentation pipeline for zero-shot human-AI coordination evaluation including turnkey experimental procedures and scale design. Through the pipeline, we engaged 130 participants in human experiments, and the results highlighted a general preference for our approach over SOTA methods across various subjective metrics. Furthermore, our objective experiments in the Overcooked environment demonstrated that our algorithm, referred to as COLE_{SV}, exceeded the performance of SOTA algorithms when coordinating with new AI agents and the human proxy model. It also demonstrated the efficient resolution of cooperative incompatibility.

Limitations. Convergence of COLE framework requires satisfying the assumption that the preference centrality of the newly generated strategy is ranked in the top k, which is controlled by an additional hyperparameter. If k is too big with a lower ranking, the COLE framework will slowly converge to the best-preferred strategy. On the other hand, if k is too small and refers to a higher ranking, each generation's assumption will not be guaranteed, which will easily cause learning failure. Besides, in our implemented algorithm COLE_{SV}, we introduce the Shapley Value as the tool and develop the Graphic Shapley Value to analyze the cooperative ability. Although we have utilized Monte Carlo permutation sampling to reduce the computational complexity, the computational complexity is still high. Therefore, we only maintain a population of 50 for the limitation of computational resources.

Future Work. Future work will focus on performing an adaptive mechanism that automatically selects a suitable value for the hyperparameter k to improve the convergence rate without promoting iterations of each update. Meanwhile, improving the efficiency of the graphic Shapley Value solver and exploring other cooperative ability evaluation solvers are important in developing the framework. Future work also includes the development of practical algorithms for more complex games except for Overcooked.

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Appendix A. Proofs of Theorem 4.4

Theorem 4.4. *Let $s_0 \in \mathcal{S}$ be the initial strategy and $s_i = \text{oracle}(s_{i-1})$ for $i \in \mathbb{N}$. Under the effective functioning of the approximated oracle as characterized by Eq. 5, we can say that the sequence $\{s_i\}$ for $i \in \mathbb{N}$ could converge to a local optimal strategy s^*, i.e., the local best-preferred strategy.*

Proof. According to the definition of the local best-preferred strategy, the local optimal strategy is the node with zero preference centrality (η). Therefore, we need to prove that the value of η will approach zero.

Let η_t denote the centrality value of the preference of the updated strategy s_t in generation t, where $0 \leq \eta \leq 1$. We first remark on the RL oracle defined in Eq. 5: $s_{n+1} = \text{oracle}(s_n, \mathcal{J}(s_n, \phi)),\ \textit{with}\ \mathcal{R}(\eta(s_{n+1})) > k$. Under the assumption of the approximated oracle functioning effectively, it follows that the preference centrality η_t associated with generated strategy s_t at generation t resides among the first k positions when arranged in ascending order of centrality values. For simplicity, We define a group g_t at generation t as the set of strategies with the lowest k preference centrality values.

Lemma A.1. *Provided that the approximated oracle is functioning effectively, the maximal preference centrality value, denoted as η_{g_t}, within group g_t is expected to exhibit a diminishing trend with each successive generation.*

Proof. In light of the approximated oracle's definition, the preference centrality value of the strategy s_t, generated at generation t, will occupy one of the first k positions when sorted in ascending order based on preference centrality values. Consequently, the initial k strategies of group g_t will undergo an update, in which the strategy ranking k-th with the highest preference centrality value is substituted by either the $(k-1)$-th strategy or the newly generation strategy, s_t. Under any given conditions, the maximum preferential value within the group will experience a reduction. □

Let η_{g_t} denote the largest preference centrality in the group g. Thus, we can derive the subsequent equation based on Lemma A.1.

$$\eta_{g_t} = \eta_{g_{t-1}} - \epsilon_{t-1}, \tag{9}$$

where ϵ_{t-1} is a positive value and $0 < \epsilon \leq \eta_{g_{t-1}}$. By further simplifying the equation, we have

$$\begin{aligned} \eta_{g_t} &= \eta_{g_{t-1}} - \epsilon_{t-1}, \\ &= \eta_{g_{t-1}} - \alpha_{t-1}\eta_{g_{t-1}}, \\ &= \beta_{t-1}\eta_{g_{t-1}}, \end{aligned} \tag{10}$$

where the second line employs $\eta_{g_{t-1}}$ to substitute the residual term, with the adjustment parameters $0 < \alpha_{t-1} \leq 1$ and $\beta_{t-1} = 1 - \alpha_{t-1}$.

Assuming that the centrality value of the preference in the initial step is $0 \leq \eta_0 \leq 1$, we can recursively calculate the following formula:

$$\begin{aligned} \eta_{g_t} &= \beta_{t-1}\eta_{g_{t-1}}, \\ &= \beta_{t-1}\beta_{t-2}\eta_{g_{t-2}}, \\ &= \cdots, \\ &= \prod_{i=0}^{t-1} \beta_i \times \eta_{g_0}. \end{aligned} \tag{11}$$

For any $\beta \in \beta_0, \cdots, \beta_{t-1}$, we have $1 > \beta \geq 0$. In addition, we set β_t as a very small positive number if $\eta_t = 0$. Furthermore, we ascertain that the coefficient β is not consistently zero. This is due to the fact that $\beta = 0$ would imply a preference centrality of zero for the strategy, which is not universally attainable within the context of the RL oracle. This very limitation underpins our rationale for introducing the approximated RL oracle. Thus, we can conclude that η_t will approach zero within the population as outlined in equation 11.

Through this proof, we have substantiated that under the effective functioning of the RL oracle as characterized by Eq. 5, the sequence s_i is progressing towards the zero of preference centrality within the population. That is, the sequence is converging to a strategy denoted by s^*, which represents a locally best-preferred solution. □

Appendix B. Proof of Corollary 4.5

Corollary 4.5. *Let $\eta : \mathcal{G}' \to \mathbb{R}^n$ be a function that maps a P-GFG to its in-degree centrality, the convergence rate of the sequence $\{s_i\}$ is Q-sublinear concerning η.*

Proof. In Theorem 4.4, we have proved that the strategies generated by the COLE framework will converge to the local best-preferred strategy. When we use the in-degree centrality function as η, the preference centrality function can be rewritten as:

$$\eta(i) = 1 - \frac{I_i}{n-1}, \tag{12}$$

where I_i is the in-degree of node i and n is the size of the strategy set $\mathcal{N}$.

Therefore, we have

$$\begin{aligned} &\lim_{t\to\infty} \frac{|\eta_{t+1} - 0|}{|\eta_t - 0|} \\ =&\lim_{t\to\infty} \frac{\eta_{t+1}}{\eta_t} \\ =&\lim_{t\to\infty} \frac{1 - \frac{I_{t+1}}{t}}{1 - \frac{I_t}{t-1}} \\ =&\lim_{t\to\infty} \frac{t-1}{t} \frac{t - I_{t+1}}{t - I_t - 1} \\ =&\lim_{t\to\infty} \frac{t - I_{t+1}}{t - I_t - 1} \\ =&1 \end{aligned} \tag{13}$$